Neuro-Symbolic Instruction Tuning for Explainable Mahjong Agents via Two-Stage Dual-LoRA

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Abstract. While recent artificial agents have achieved superhuman performance in Riichi Mahjong, they predominantly operate as opaque black boxes, limiting their pedagogical value for human players. To bridge this gap, we introduce NeSy-Mahjong, an interpretable neuro-symbolic system designed for *Nanikiru* problems in Riichi Mahjong. Addressing the challenges of catastrophic forgetting and logical hallucination in large language models, our approach employs a decoupled two-stage dual-LoRA architecture. We first align a decision adapter with professional intuition, then train a separate reasoning adapter grounded by a symbolic calculator, which injects objective metrics (e.g., *Shanten*, *Ukeire*) into the generation context. Extensive experiments on a dataset derived from high-level Tenhou games demonstrate that our sub-10B open-source models significantly outperform state-of-the-art commercial models including DeepSeek-V3 and Qwen-Plus in both decision accuracy and explanation quality.

Keywords: Riichi Mahjong · Neuro-Symbolic AI · Explainable AI · Large Language Models · Instruction Tuning.

1 Introduction

The pursuit of superhuman performance in games has long been a grand challenge and a pivotal testing ground for artificial intelligence (AI). Over the past decade, the field has witnessed remarkable breakthroughs, with progress evolving from mastering perfect-information games such as Go [29] to conquering complex imperfect-information domains including Diplomacy [3] and Werewolf [36]. In these arenas, AI agents have demonstrated sophisticated capabilities, including mastering intricate strategies, executing bluffs, and engaging in long-term planning.

Among these, Japanese Riichi Mahjong stands out as a particularly challenging frontier due to its unique combination of multi-player competition, stochastic tile distribution, and complex scoring rules. Recent reinforcement learning

(RL) agents, most notably Suphx [21] and various commercial engines including NAGA [7] and Mortal [8], have achieved top-tier levels on the game platform Tenhou [33], surpassing average human professionals in terms of win rates and stable performance.

However, while these state-of-the-art (SOTA) systems have achieved remarkable playing strength, they primarily operate as black boxes, outputting decisions with probability distributions or value estimates without articulating the underlying strategic rationale. For the vast community of players, especially novices, such numerical outputs offer limited pedagogical value. In Riichi Mahjong, discard decisions require balancing offensive progress, defensive risk, and long-term considerations under imperfect information. Bridging this gap therefore calls for explainable AI [12, 25] methods that translate implicit strategic judgments into explicit language.

Motivated by this challenge, this paper shifts the research focus from pursuing incremental gains in playing strength to building an interpretable Mahjong coach. Specifically, we narrow our scope to the *Nanikiru* (what-to-discard) problems, which constitute the most frequent and fundamental decision-making process in Riichi Mahjong and encapsulate the core trade-off between efficiency and safety. While full-game gameplay involves additional mechanics such as calling (*Chi*/*Pon*/*Kan*), discard decisions remain the primary cognitive bottleneck for learners operating under imperfect information. Our objective is thus to generate coherent and pedagogically sound natural language explanations for these critical decisions.

To achieve this, we introduce NeSy-Mahjong, an interpretable neuro-symbolic Riichi Mahjong system designed for the *Nanikiru* problem. To address common limitations of standard supervised fine-tuning [26], such as catastrophic forgetting and factual hallucinations, we employ a two-stage fine-tuning strategy using parameter-efficient low-rank adaptation (LoRA) adapters [14]. In the first stage, we align a lightweight decision expert (LoRA-A) with high-level human intuition. In the second stage, we decouple reasoning by freezing the decision module and training a separate reasoning expert (LoRA-B) via teacher-student distillation. Crucially, we integrate a deterministic symbolic calculator that injects objective Mahjong metrics (e.g., *Shanten*, *Ukeire*, *Genbutsu*) into the generation context, ensuring that explanations are not only linguistically coherent but also mathematically grounded, thereby mitigating logical hallucinations [17].

The main contributions of this paper are summarized as follows:

1. We propose NeSy-Mahjong, a neuro-symbolic framework with a decoupled two-stage dual-LoRA architecture that separates decision alignment from linguistic reasoning, alleviating catastrophic forgetting.
2. We integrate a deterministic symbolic calculator to inject objective Mahjong metrics into the reasoning process, significantly improving factual correctness and logical consistency.
3. We construct and release a comprehensive benchmark dataset derived from high-level Tenhou gameplay and Mahjong Soul community votes, addressing

the scarcity of reasoning-oriented Mahjong data. The dataset and implementation are publicly available at [9].

4. Experiments show that our sub-10B-scale models substantially outperform vanilla baselines and commercial large language models (LLMs) in both decision accuracy and explanation quality, as validated by quantitative metrics and human evaluation.

2 Related Work

2.1 AI for Mahjong Gameplay

As a typical representative of imperfect-information scenarios, Riichi Mahjong has seen remarkable progress with RL-based systems such as Suphx [21] and commercial engines [4, 7, 8]. In the recent IJCAI 2025 Mahjong AI Competition [16], the winning team achieved top-tier performance also using an RL-based approach. Moreover, due to the rapid development of LLMs, recent works have explored their potential in card games. For instance, Wang et al. [34] evaluated LLMs across eight card games, including Riichi Mahjong, and showed that fine-tuned small-scale models can outperform closed-source SOTA models in decision-making.

However, a critical limitation across these systems is their black-box nature. While they excel at decision-making, they lack the capacity to articulate their reasoning. Although Mxplainer [22] improves interpretability, it mainly offers parameter-level hints such as feature weights, rather than natural language explanations. This leaves a critical gap for systems designed to convey strategic reasoning to human players.

2.2 Neuro-Symbolic Grounding

LLMs frequently suffer from hallucinations in tasks requiring rigorous logic or arithmetic, often failing on basic numerical comparisons or strict rule adherence [23, 35]. In *Nanikiru* problems of Riichi Mahjong, this fragility is fatal, because concepts such as *Shanten* (moves to win) and *Ukeire* (tile acceptance) require strictly precise calculation rather than probabilistic generation.

To this end, neuro-symbolic AI [10] offers a solution by combining the generative flexibility of neural networks with the logical correctness of symbolic systems. Recent works [28, 18] have demonstrated that injecting external knowledge into LLM contexts improves factual accuracy. Building on this, we integrate a deterministic symbolic calculator encoding critical game mechanics in *Nanikiru* problems to ground explanations in mathematically accurate game states.

2.3 Parameter-Efficient Fine-Tuning

To address the substantial computational costs of adapting LLMs, parameter-efficient fine-tuning (PEFT) [13] has emerged as a standard paradigm. Among

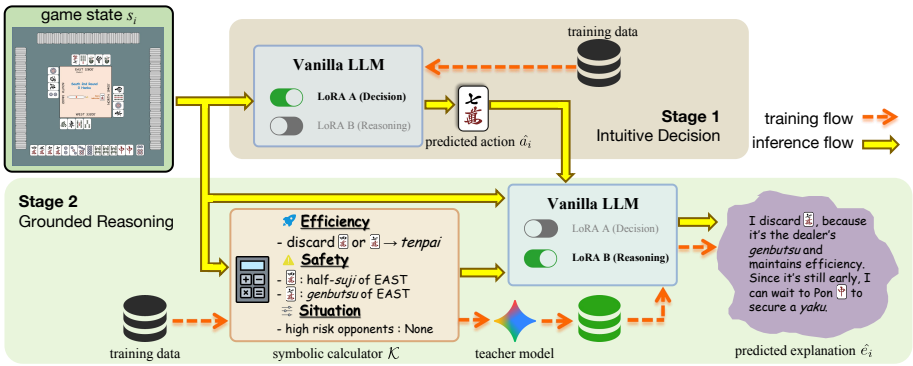

Fig. 1. The Architecture of NeSy-Mahjong.

these methods, LoRA [14] has gained prominence for its efficiency. LoRA freezes the pre-trained model weights and injects trainable rank decomposition matrices into the Transformer layers, allowing the model to adapt to downstream tasks with a minimal increase in trainable parameters.

However, applying such fine-tuning in multi-objective scenarios presents challenges. Prior research [20, 24] indicates that sequentially optimizing for disparate tasks, such as decision-making and reasoning, within a shared parameter space often induces task interference or catastrophic forgetting. This highlights the limitations of standard single-adapter approaches when an agent is required to master conflicting competencies simultaneously. To address this, we adopt a dual-LoRA architecture, detailed in Section 3.3.

3 Methodology

We propose NeSy-Mahjong, a decoupled neuro-symbolic framework illustrated in Figure 1, which integrates symbolic computation with offline teacher distillation under a dual-LoRA architecture. This section presents details of the methods.

3.1 Problem Formulation

Nanikiru problems involve imperfect information and can be modeled as a partially observable Markov decision process [19]. At each turn, the ego-centric player observes a partial game state s, consisting of the private hand, publicly visible information such as discards and open melds, and round-level context. The task then focuses on the discard decision, where the agent selects a tile a from its hand under incomplete information. This decision involves inherent trade-offs between efficiency and safety, and multiple actions may be considered reasonable by human experts. Accordingly, we formulate a *Nanikiru* problem as a policy alignment problem rather than a single-solution optimization task.

Let $\mathcal{D} = \{(s_i, a_i, e_i)\}_{i=1}^{N}$ denote a dataset of expert demonstrations, where a_i is the chosen discard and e_i is the corresponding natural-language explanation. Due to stylistic and strategic variations among experts, actions for a given state can be viewed as samples from a multi-modal distribution $P_{\text{expert}}(a \mid s)$.

Our objective is to train an LLM parameterized by θ to approximate this expert policy while generating coherent explanations consistent with the selected action. We define the objective as maximizing the joint probability of the decision–explanation pair as:

$$\max_{\theta} \sum_{(s,a,e)\in\mathcal{D}} \log P_\theta(a, e \mid s). \tag{1}$$

3.2 Neuro-Symbolic Knowledge Injection

To bridge the gap between the implicit numerical features of game states and the explicit natural language reasoning required for explanation generation, we integrate a deterministic symbolic calculator [18, 28] into our framework. This calculator, denoted as $\mathcal{K}$, maps a raw game state s into a structured set of objective metrics $k = \mathcal{K}(s)$. These metrics ensure that the LLM reasoning is grounded in the rigorous mathematics of *Nanikiru* problems.

The symbolic calculator explicitly models three critical dimensions, including efficiency, safety, and situation. Among these, efficiency metrics are formally defined and formulated in this section. In contrast, safety and situational metrics are largely heuristic in nature, and we briefly describe them here and provide their formal definitions in the online supplementary document provided in [9].

Efficiency A fundamental objective in *Nanikiru* problems is to advance the hand toward completion efficiently, which primarily involves the following two metrics.

1. **Shanten ($\mathcal{S}$):** the minimum edit distance to a valid *Tenpai* (ready-to-win) state. Let Ω_{valid} be the universe of valid hands and $\mathcal{T} \subset \Omega_{\text{valid}}$ be the subset of Tenpai hands. The *Shanten* of hand H is:

$$\mathcal{S}(H) = \min_{H'\in\mathcal{T}} \text{dist}(H, H'), \tag{2}$$

   where dist$(\cdot)$ represents tile substitutions. $\mathcal{S}(H) = 0$ implies H is in *Tenpai*.
2. **Visible Ukeire ($\mathcal{U}_{\text{vis}}$):** the remaining number of drawable tiles that reduce the *Shanten* number under observable tile constraints. We define the set of improving tiles as:

$$T_{\text{imp}} = \{\, t \mid \exists x \in H,\ \mathcal{S}(H \setminus \{x\} \cup \{t\}) < \mathcal{S}(H) \}\,. \tag{3}$$

   The visible Ukeire is then computed as

$$\mathcal{U}_{\text{vis}}(H) = \sum_{t\in T_{\text{imp}}} \max\left(0, 4 - N_{\text{vis}}(t)\right), \tag{4}$$

   where $N_{\text{vis}}(t)$ denotes the total number of observable instances of tile t.

Safety Efficiency-oriented discards may still incur substantial defensive risk. Safety metrics evaluate the risk of a candidate discard using only observable information from opponents. Our symbolic calculator identifies one provably safe pattern, namely *Genbutsu*, as well as several widely adopted heuristic indicators, including *Suji* structures and tile exhaustion. Together, these indicators reflect different levels of discard safety based solely on observable information.

Situation Optimal decisions in *Nanikiru* problems depend on the global game context. Situational metrics capture this context and guide the prioritization between efficiency and safety. We estimate the threat level of each opponent using a dynamic heuristic score, which aggregates signals from open melds, discard progression, and *Dora* visibility. The resulting score is discretized into several threat levels, providing coarse-grained guidance for balancing offensive efficiency and defensive safety.

3.3 Data Construction via Neuro-Symbolic Distillation

Constructing a high-quality instruction tuning dataset for *Nanikiru* problems is non-trivial because of the scarcity of reasoning data available from gameplay or community platforms. To overcome this, we employ a teacher-student distillation strategy, leveraging the symbolic knowledge defined in Section 3.2 to ground the reasoning process of the teacher model.

We utilize Gemini-2.0-Flash [31], a strong proprietary model, as the teacher. To ensure the quality of reasoning data generated by the teacher model, we augment the prompt with the objective metrics $k = \mathcal{K}(s)$ calculated by our symbolic calculator. The generation process can be viewed as:

$$e \sim P_{\text{Teacher}}(e \mid s \oplus k \oplus a), \tag{5}$$

where $\oplus$ denotes textual concatenation. Here, the teacher is instructed to treat the symbolic metrics k as explicit expert constraints and articulate the logic connecting the state s to the ground-truth action a.

This symbolic-augmented prompting ensures that the generated explanations are factually consistent with the mathematical reality of the game. For instance, if the calculator identifies a tile as *Genbutsu*, the teacher is forced to incorporate this safety guarantee into its reasoning. The resulting dataset $\mathcal{D}_{\text{reason}}$ serves as the training set for reasoning.

3.4 NeSy-Mahjong Architecture

To reconcile the conflicting objectives of precise strategic decision-making and fluent linguistic reasoning, we propose a decoupled dual-LoRA architecture that separates these capabilities into distinct parameter spaces. The framework adopts a two-stage training strategy with dynamic adapter switching at inference time.

Stage 1: Intuitive Decision (LoRA-A) In the first stage, we align a frozen pre-trained base model $\mathcal{M}_\Theta$ with expert-level strategic intuition. We introduce a low-rank adapter $\Delta\theta_A$ (LoRA-A) dedicated to decision-making and train it on expert state-action pairs $\mathcal{D}_{\text{decision}} = \{(s_i, a_i)\}$. Formally, this stage can be viewed as optimizing a conditional likelihood objective:

$$\mathcal{L}_{\text{stage1}} = -\sum_{(s,a)\in\mathcal{D}_{\text{decision}}} \log P(a \mid s; \Theta, \Delta\theta_A). \tag{6}$$

To avoid verbose hallucination and reduce latency, the model is trained to produce a minimal structured action output rather than free-form reasoning. The resulting adapter $\Delta\theta_A^*$ captures the implicit expert policy.

Stage 2: Grounded Reasoning (LoRA-B) The second stage focuses on explanation generation while preserving the decision boundary learned in Stage 1. We freeze $\Delta\theta_A^*$ and introduce a separate reasoning adapter $\Delta\theta_B$ (LoRA-B), which is trained on the synthetic dataset $\mathcal{D}_{\text{reason}} = \{(s_i, a_i, k_i, e_i)\}$ constructed in Section 3.3. The training objective of this stage is defined analogously as:

$$\mathcal{L}_{\text{stage2}} = -\sum_{(s,a,k,e)\in\mathcal{D}_{\text{reason}}} \log P(e \mid s \oplus k \oplus a; \Theta, \Delta\theta_B). \tag{7}$$

By explicitly conditioning on symbolic metrics k, LoRA-B learns to generate grounded explanations rather than hallucinated reasoning.

Inference: Dynamic Adapter Switching At inference time, the system operates in two phases. First, LoRA-A is activated to predict the discard:

$$\hat{a} = \arg\max_{a'} P(a' \mid s; \Theta, \Delta\theta_A^*). \tag{8}$$

The symbolic calculator then computes $k = \mathcal{K}(s)$, after which LoRA-A is deactivated and LoRA-B is activated to generate the explanation:

$$\hat{e} \sim P(e \mid s \oplus k \oplus \hat{a}; \Theta, \Delta\theta_B^*). \tag{9}$$

This decoupled inference pipeline preserves decision accuracy while enabling neuro-symbolically grounded explanations.

4 Experiments

In this section, we evaluate NeSy-Mahjong on decision accuracy, reasoning quality, and generalization. Additional case studies are provided in the online supplementary document [9].

4.1 Experimental Setup

Data Collection and Preprocessing To construct a dataset for *Nanikiru* problems, we utilize real-world gameplay logs released by the Tenhou platform[4] for major experiments. Specifically, we select logs where all four players are ranked 7-*dan* or above. For each match, we select a top-2 player and extract their discard actions to form samples. Notably, post-*Riichi* turns of the player are excluded, as discards in this state are fixed and non-volitional. Samples are uniformly drawn across round phases (Early/Mid/Late) and strategic contexts (Offense/Defense), where the latter is determined by whether any opponent has declared *Riichi*. Detailed preprocessing steps are provided in the online supplementary document [9]. Eventually, we curated 44,034 training and 4,416 validation samples for Stage 1, and 5,544 samples for Stage 2 training, as well as 4,896 samples for testing. Additionally, we collected 20 crowdsourced *Nanikiru* problems from Mahjong Soul's official Bilibili account[5]. Each problem attracted over 2,000 participants. We use this benchmark to evaluate generalization.

Implementation Details. We used PyTorch 2.9 and the HuggingFace Transformers 4.49 library for experiments. Models were fine-tuned on NVIDIA RTX A6000 (48GB) GPUs with CUDA 12.2 support. To implement the decoupled architecture, we managed the dual-stage adapters using the PEFT library and optimized the training pipeline with HuggingFace Accelerate. We provide more implementation details, including full hyper-parameters, in our online supplementary document [9].

Baselines We evaluate NeSy-Mahjong against two types of baselines. On one hand, we use the vanilla versions of our sub-10B-scale open-source backbones: DeepSeek-LLM-7B-Chat [5], Llama-3.1-8B-Instruct [11], Qwen2.5-7B-Instruct [27], and Yi-6B-Chat [1]. On the other hand, in the absence of task-specific prior work, we also compare against API-based closed-source LLMs, including Qwen-Plus [2], DeepSeek-V3 [6], GLM-4 [30], Moonshot-Kimi [32], and Gemini-2.0-Flash [31]. The exact prompt templates used for all models are detailed in the online supplementary document [9].

Research Questions We evaluate NeSy-Mahjong through the following research questions (RQs):

RQ1: Decision Efficacy. How accurately does NeSy-Mahjong make discard decisions in *Nanikiru* settings?

RQ2: Reasoning Quality. How coherent, factual, and tactically grounded are the explanations generated by NeSy-Mahjong?

RQ3: Mechanisms and Ablation. How much do the dual-stage training scheme and neuro-symbolic knowledge injection contribute to the performance?

[4] `https://tenhou.net/sc/raw/` (Accessed: 2026-04-20).
[5] `https://space.bilibili.com/353240497` (Accessed: 2026-04-20).

4.2 Experimental Results

Results for RQ1 To evaluate discard decision accuracy in *Nanikiru* problems, we adopt a human-grounded evaluation using player choices in the test set as ground truth. For the models with our methods, we use a small beam search to obtain Top-3 predictions, while vanilla and API-based models are evaluated with a single output due to decoding constraints. Results are shown in Table 1.

Table 1. Overall decision accuracy on the Tenhou benchmark. Models fine-tuned with our methods are denoted as "NeSy-Backbone" (e.g., NeSy-Qwen). **Bold** indicates the best performance within each column.

Model	**Top-1 Acc**	**Top-3 Acc**
DeepSeek-LLM-7B	3.3%	–
Llama-3.1-8B	3.0%	–
Qwen2.5-7B	8.8%	–
Yi-1.5-6B	3.6%	–
Qwen-Plus	22.5%	–
DeepSeek-V3	32.1%	–
GLM-4	25.4%	–
Moonshot-Kimi	24.1%	–
Gemini-2.0-Flash	19.9%	–
NeSy-DeepSeek	65.4%	**88.7%**
NeSy-Llama	66.5%	88.4%
NeSy-Qwen	**67.0%**	87.0%
NeSy-Yi	66.2%	86.5%

As shown in Table 1, vanilla models perform extremely poorly, with Top-1 accuracy below 10%, which is merely comparable to random tile selection from a typical 14-tile hand. Case inspection further shows that vanilla models often output tiles not even present in the player's hand, indicating a lack of basic rule understanding rather than insufficient reasoning. Commercial API-based LLMs perform better but still fall short: even the strongest model (DeepSeek-V3) reaches only 32.1% Top-1 accuracy. In contrast, the models with our methods achieve a substantial performance gain, with Top-1 accuracy rising to 65–67% across backbones, demonstrating clear superiority in decision-making efficacy. For context, the landmark SOTA Suphx [21] reported 76.7% accuracy. Given our lighter training scale and stricter data filtering that excludes post-*Riichi* turns, the results indicate strong competitiveness.

Furthermore, Figure 2 presents a fine-grained performance breakdown across different game phases. The results indicate strong stability across all scenarios without notable degradation. We observe that Top-1 accuracy exhibits an

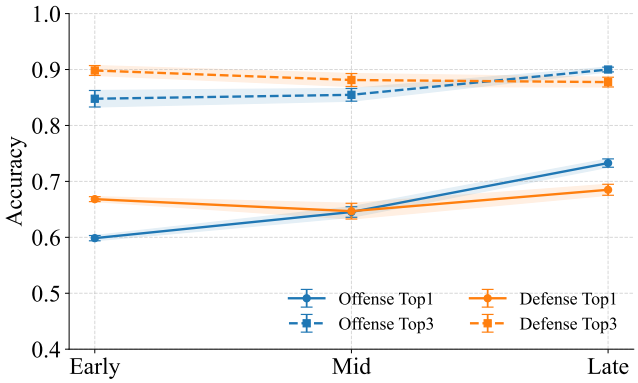

Fig. 2. Fine-grained decision accuracy across game phases (Early/Mid/Late) and strategic contexts (Offense/Defense). The plots report the average performance of NeSy-Mahjong based on four LLMs, with error bars denoting the standard deviation.

upward trend from early to late stages, while the discrepancy between Top-1 and Top-3 accuracy generally narrows over time. These converging trends collectively suggest that opening discards are intrinsically diverse with multiple valid options, whereas late-stage decisions become increasingly constrained and deterministic, resulting in higher predictability.

Results for RQ2 To evaluate the reasoning quality of each model, we adopt an LLM-as-a-Judge strategy using GPT-4o, whose human-aligned scoring behavior and robustness have been validated [15]. Moreover, to control evaluation cost while preserving coverage, we sample 40 instances from each combination of game phase (Early/Mid/Late) and strategic context (Offense/Defense). For each *Nanikiru* problem, the judge receives the full game state, the model-generated explanation, and symbolic reference metrics, and scores the explanation along three dimensions on a 1–10 scale: logical coherence, factual precision, and strategic soundness. The results are shown in Table 2.

In Table 2, across all metrics, our methods substantially outperform both vanilla and API-based systems. Vanilla models, which lack effective rule grounding, score below 3.0 in all dimensions. API-based LLMs exhibit strong linguistic coherence but noticeably weaker factual reliability, with frequent hallucinations on safety claims and tile-efficiency metrics.

In contrast, our methods achieve near-expert performance across all evaluation criteria, with NeSy-Llama consistently obtaining the highest scores. These gains confirm that symbolic grounding effectively anchors the models to objective game-state computations, preventing hallucinated statistics or unsafe tile judgments. Moreover, our methods also obtain the best Strategy scores, indicating that they not only avoid factual mistakes but also synthesize coherent, tactically sound reasoning aligned with professional Mahjong principles.

Table 2. Explanation quality evaluated by GPT-4o. We report average scores for logical coherence (Logic), factual precision (Precision), and strategic soundness (Strategy). **Bold** indicates the best performance within each column.

Model	Logic	Precision	Strategy
Llama-3.1-8B	2.26	1.38	1.40
DeepSeek-LLM-7B	2.09	1.46	1.44
Yi-1.5-6B	2.37	1.61	1.60
Qwen2.5-7B	2.93	2.14	2.21
Qwen-Plus	7.61	7.38	7.07
DeepSeek-V3	7.78	7.44	7.35
GLM-4	7.05	6.72	6.47
Moonshot-Kimi	7.32	7.19	6.82
Gemini-2.0-Flash	7.18	6.93	6.51
NeSy-DeepSeek	8.09	8.39	7.75
NeSy-Llama	**8.12**	**8.48**	**7.82**
NeSy-Qwen	8.08	8.43	7.81
NeSy-Yi	8.01	8.33	7.78

Results for RQ3 To understand the contribution of each component in NeSy-Mahjong, we conduct ablation studies focusing on the dual-stage LoRA architecture, and the neuro-symbolic knowledge injection. We create two types of variants by removing the neuro-symbolic grounding at inference, and by sequentially training a single LoRA. Table 3 shows the performance of these variants.

Table 3 shows that removing neuro-symbolic grounding consistently reduces reasoning quality, confirming its necessity for stable Stage-2 explanations. In contrast, decision accuracy remains unchanged, as expected from its design. Moreover, the single-LoRA variants exhibit much larger performance degradation, especially for Stage-1 performance of DeepSeek and Qwen, indicating strong interference between decision and reasoning objectives. This suggests these backbones are more prone to catastrophic forgetting when parameters are not decoupled. By comparison, Yi shows a more moderate decline, implying greater robustness.

Overall, the results validate the importance of the dual-stage architecture for preserving decision performance while supporting high-quality reasoning.

Human Alignment and Generalization To evaluate whether NeSy-Mahjong generalizes to out-of-distribution (OOD) scenarios, we use selected *Nanikiru* problems released by Mahjong Soul. Specifically, we compare the beam-search top-3 candidates against human vote distributions to assess alignment with human preferences. We report five alignment metrics in Table 4: Top-1 Accuracy, Soft Match (model top-1 appears in the human top-3 set), Precision@3 (top-k

Table 3. Ablation variants of NeSy-Mahjong. "w/o NS" removes neuro-symbolic grounding at inference, affecting only Stage 2 reasoning. "Single-LoRA" applies both training stages to a single LoRA module.

Model Variant	Top-1 Acc	Logic	Precision	Strategy
NeSy-DeepSeek w/o NS	-	6.38	6.27	5.75
NeSy-Llama w/o NS	-	6.72	6.67	6.08
NeSy-Qwen w/o NS	-	6.59	6.57	6.05
NeSy-Yi w/o NS	-	6.30	6.16	5.62
NeSy-DeepSeek Single-LoRA	11.8%	6.79	6.64	6.10
NeSy-Llama Single-LoRA	43.4%	5.24	5.41	5.05
NeSy-Qwen Single-LoRA	17.7%	6.71	6.80	6.20
NeSy-Yi Single-LoRA	57.5%	7.93	8.20	7.53

overlap), Total Probability Mass (the human probability assigned to the model's top-3 options), and NDCG@3 (ranking consistency with human preferences).

Table 4. Human alignment on 20 Mahjong Soul *Nanikiru* problems. We report Top-1 Accuracy (Acc), Soft Match (Soft), Precision@3 (P@3), Total Probability Mass (TPM), and NDCG@3.

Model	Acc	Soft	P@3	TPM	NDCG
NeSy-DeepSeek	0.400	0.750	0.450	0.526	0.552
NeSy-Llama	0.250	0.550	0.500	0.539	0.524
NeSy-Qwen	0.400	0.750	0.433	0.513	0.545
NeSy-Yi	0.400	0.600	0.467	0.555	0.552

For reference, across these problems, the human top choice receives on average 43.85% of the votes, reflecting substantial disagreement even among human players. Remarkably, our models achieve up to 40% Top-1 Accuracy, approaching the human agreement ceiling and demonstrating strong OOD generalization.

To further assess explanation quality, we conduct a human preference study involving four experienced Mahjong players with over three years of experience each. To keep the questionnaire manageable, we only compare two leading API-based models (DeepSeek-V3, Qwen-Plus) and two best-performing variants of our method (NeSy-Llama, NeSy-Qwen). For each of the 20 *Nanikiru* problems, participants select the most convincing explanation among the four candidates.

Across the 20 problems, NeSy-Llama receives the highest number of human votes, followed by NeSy-Qwen, while the two API-based models obtain substantially fewer preferences. This ranking mirrors the earlier GPT-4o evaluation, where NeSy-Llama achieved the strongest overall scores. The consistency be-

Table 5. Human preference evaluation on 20 *Nanikiru* problems. We report the number of votes obtained by each model out of a total of $20 \times 4 = 80$ votes.

Model	Human Preference
DeepSeek-V3	17
Qwen-Plus	9
NeSy-Llama	**30**
NeSy-Qwen	24

tween human judgments and LLM-as-a-judge results suggests that our neuro-symbolically grounded models not only produce objectively accurate reasoning but also generate explanations that human players find more convincing and aligned with expert intuition.

5 Conclusion

In this paper, we presented NeSy-Mahjong, a framework that integrates intuitive neural decision-making with symbolic reasoning. By combining a symbolic calculator with a decoupled dual-LoRA architecture, we mitigate factual hallucination and catastrophic forgetting commonly observed in standard LLM fine-tuning. Our results demonstrate a compelling finding: small-scale, domain-specialized models, when grounded with explicit symbolic knowledge, can surpass general-purpose commercial giants in complex strategic reasoning tasks.

Future work will extend this paradigm to full-game scenarios, incorporating auxiliary mechanics such as calling. We also plan to explore multi-agent architectures to achieve more comprehensive and interpretable modeling of Mahjong gameplay, enabling coherent explanations across diverse in-game interactions.

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